

PSDA · frame + one-variable data

41 items · two PSDA families only · untimed

1–11 Family A · official · frame / generalize / MoE / experiment

12–25 Family B · official · frequency / box plot / two data sets

26–35 Family A · repo originals (Sets 06 / 07 + desk)

36–41 Family B · repo originals (Sets 03 / 04 / 06)

Official pages are the Hard exclude-active bank. Originals are not
College Board stems. Do not re-sit Bluebook 4 / 5 / 6 / 8 / 9 / 11.

Question ID: 85939da5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Hard ■■■■■

Question

Texting behavior	Talks on cell phone daily	Does not talk on cell phone daily	Total
Light	110	146	256
Medium	139	164	303
Heavy	166	74	240
Total	415	384	799

In a study of cell phone use, 799 randomly selected US teens were asked how often they talked on a cell phone and about their texting behavior. The data are summarized in the table above. Based on the data from the study, an estimate of the percent of US teens who are heavy texters is 30% and the associated margin of error is 3%. Which of the following is a correct statement based on the given margin of error?

Answer

- A. Approximately 3% of the teens in the study who are classified as heavy texters are not really heavy texters.
- B. It is not possible that the percent of all US teens who are heavy texters is less than 27%.
- C. The percent of all US teens who are heavy texters is 33%.
- D. It is doubtful that the percent of all US teens who are heavy texters is 35%.

Question ID: 1ea09200

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Evaluating statistical claims: Observational studies and experiments	Hard ■■■■■

Question

A sample of 40 fourth-grade students was selected at random from a certain school. The 40 students completed a survey about the morning announcements, and 32 thought the announcements were helpful. Which of the following is the largest population to which the results of the survey can be applied?

Answer

- A. The 40 students who were surveyed
- B. All fourth-grade students at the school
- C. All students at the school
- D. All fourth-grade students in the county in which the school is located

Question ID: 308084c5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Hard

Question

Sample	Percent in favor	Margin of error
A	52%	4.2%
B	48%	1.6%

The results of two random samples of votes for a proposition are shown above. The samples were selected from the same population, and the margins of error were calculated using the same method. Which of the following is the most appropriate reason that the margin of error for sample A is greater than the margin of error for sample B?

Answer

- A. Sample A had a smaller number of votes that could not be recorded.
- B. Sample A had a higher percent of favorable responses.
- C. Sample A had a larger sample size.
- D. Sample A had a smaller sample size.

Question ID: 7d68096f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Evaluating statistical claims: Observational studies and experiments	Hard ■■■■■

Question

A trivia tournament organizer wanted to study the relationship between the number of points a team scores in a trivia round and the number of hours that a team practices each week. For the study, the organizer selected **55** teams at random from all trivia teams in a certain tournament. The table displays the information for the **40** teams in the sample that practiced for at least **3** hours per week.

Hours practiced	Number of points per round		
	6 to 13 points	14 or more points	Total
3 to 5 hours	6	4	10
More than 5 hours	4	26	30
Total	10	30	40

Which of the following is the largest population to which the results of the study can be generalized?

Answer

- A. All trivia teams in the tournament that scored **14** or more points in the round
- B. The **55** trivia teams in the sample
- C. The **40** trivia teams in the sample that practiced for at least **3** hours per week
- D. All trivia teams in the tournament

Question ID: 7ce2830a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Evaluating statistical claims: Observational studies and experiments	Hard ■■■■■

Question

A psychologist designed and conducted a study to determine whether playing a certain educational game increases middle school students' accuracy in adding fractions. For the study, the psychologist chose a random sample of 35 students from all of the students at one of the middle schools in a large city. The psychologist found that students who played the game showed significant improvement in accuracy when adding fractions. What is the largest group to which the results of the study can be generalized?

Answer

- A. The 35 students in the sample
- B. All students at the school
- C. All middle school students in the city
- D. All students in the city

Question ID: b695711c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Evaluating statistical claims: Observational studies and experiments	Hard ■■■■■

Question

A statistician conducted a study to investigate the relationship between the average number of minutes piano students practiced each week and the percentage of correct notes the students played during a recital. For the study, the statistician selected a sample of **85** piano students at random from all the piano students in a certain region who were in their second year with their conservatory. The table shows the information for **75** piano students in this sample who practiced at least **50** minutes on average each week.

	Percentage of correct notes the students played during a recital			
Average number of minutes practiced each week	Less than 50%	50% to 80%	Greater than 80%	Total
50 to 150	7	28	5	40
151 or more	3	11	21	35
Total	10	39	26	75

Which of the following is the largest population to which the results of the study can be generalized?

Answer

- A. All piano students who were in their second year with their conservatory
- B. All piano students in the region
- C. All piano students in the region who were in their second year with their conservatory
- D. The **75** piano students in the sample who practiced at least **50** minutes on average each week

Question ID: aa43b41f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Evaluating statistical claims: Observational studies and experiments	Hard ■■■■■

Question

Near the end of a US cable news show, the host invited viewers to respond to a poll on the show's website that asked, "Do you support the new federal policy discussed during the show?" At the end of the show, the host reported that 28% responded "Yes," and 70% responded "No." Which of the following best explains why the results are unlikely to represent the sentiments of the population of the United States?

Answer

- A. The percentages do not add up to 100%, so any possible conclusions from the poll are invalid.
- B. Those who responded to the poll were not a random sample of the population of the United States.
- C. There were not 50% "Yes" responses and 50% "No" responses.
- D. The show did not allow viewers enough time to respond to the poll.

Question ID: 4a422e3e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Evaluating statistical claims: Observational studies and experiments	Hard ■■■■■

Question

To determine the mean number of children per household in a community, Tabitha surveyed 20 families at a playground. For the 20 families surveyed, the mean number of children per household was 2.4. Which of the following statements must be true?

Answer

- A. The mean number of children per household in the community is 2.4.
- B. A determination about the mean number of children per household in the community should not be made because the sample size is too small.
- C. The sampling method is flawed and may produce a biased estimate of the mean number of children per household in the community.
- D. The sampling method is not flawed and is likely to produce an unbiased estimate of the mean number of children per household in the community.

Question ID: 9ba3e283

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Hard ■■■■■

Question

In State X, Mr. Camp's eighth-grade class consisting of 26 students was surveyed and 34.6 percent of the students reported that they had at least two siblings. The average eighth-grade class size in the state is 26. If the students in Mr. Camp's class are representative of students in the state's eighth-grade classes and there are 1,800 eighth-grade classes in the state, which of the following best estimates the number of eighth-grade students in the state who have fewer than two siblings?

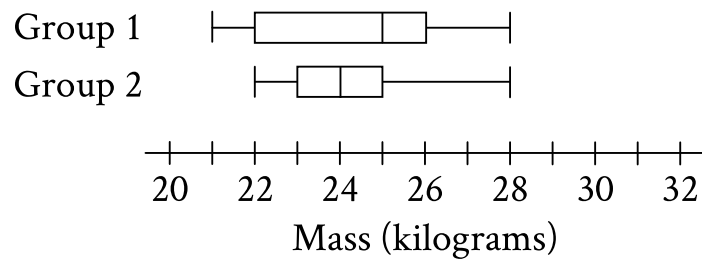
Answer

- A. 16,200
- B. 23,400
- C. 30,600
- D. 46,800

Question ID: d3b9c8d8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question



The box plots summarize the masses, in kilograms, of two groups of gazelles. Based on the box plots, which of the following statements must be true?

Answer

- A. The mean mass of group 1 is greater than the mean mass of group 2.
- B. The mean mass of group 1 is less than the mean mass of group 2.
- C. The median mass of group 1 is greater than the median mass of group 2.
- D. The median mass of group 1 is less than the median mass of group 2.

Question ID: f6c18a66

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

For data set A, the table summarizes the distribution of the number of deliveries received by an office each day during a period of **11** days.

Deliveries	Days
0	2
3	2
4	2
5	2
6	1
7	1
13	1

The data value **13** was recorded in error and is removed from data set A to create data set B, which consists of the remaining **10** data values. Which statement best compares the median of data set A and the median of data set B?

Answer

- A. The median of data set B is less than the median of data set A.
- B. The median of data set B is greater than the median of data set A.
- C. The median of data set B is equal to the median of data set A.
- D. There is not enough information to compare the medians of the two data sets.

Question ID: 190be2fc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard ■■■■■

Question

Data set A consists of 10 positive integers less than 60. The list shown gives 9 of the integers from data set A.

43, 45, 44, 43, 38, 39, 40, 46, 40

The mean of these 9 integers is 42. If the mean of data set A is an integer that is greater than 42, what is the value of the largest integer from data set A?

Question ID: db88a933

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

Data Set Q

Value	Frequency
a	110
b	30
c	0

Data Set R

Value	Frequency
a	0
b	3
c	11

Data sets Q and R are summarized in the tables shown, where a , b , and c are positive, consecutive integers and $a < b < c$. Which statement comparing the means of the data sets is true?

Answer

- A. The mean of data set R is $\frac{2(11)}{14}$ less than the mean of data set Q.
- B. The mean of data set R is $\frac{2(11)}{14}$ greater than the mean of data set Q.
- C. The mean of data set Q is **10** times the mean of data set R.
- D. The means of the two data sets are equal.

Question ID: c178d4da

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard ■■■■■

Question

Value	Data set A frequency	Data set B frequency
30	2	9
34	4	7
38	5	5
42	7	4
46	9	2

Data set A and data set B each consist of **27** values. The table shows the frequencies of the values for each data set. Which of the following statements best compares the means of the two data sets?

Answer

- A. The mean of data set A is greater than the mean of data set B.
- B. The mean of data set A is less than the mean of data set B.
- C. The mean of data set A is equal to the mean of data set B.
- D. There is not enough information to compare the means of the data sets.

Question ID: 457d2f2c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard ■■■■■

Question

A data set of 27 different numbers has a mean of 33 and a median of 33. A new data set is created by adding 7 to each number in the original data set that is greater than the median and subtracting 7 from each number in the original data set that is less than the median. Which of the following measures does NOT have the same value in both the original and new data sets?

Answer

- A. Median
- B. Mean
- C. Sum of the numbers
- D. Standard deviation

Question ID: 1142af44

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

Value	Frequency
1	a
2	2a
3	3a
4	2a
5	a

The frequency distribution above summarizes a set of data, where a is a positive integer. How much greater is the mean of the set of data than the median?

Answer

- A. 0
B. 1
C. 2
D. 3

Question ID: 1e8ccffd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard ■■■■■

Question

The mean score of 8 players in a basketball game was 14.5 points. If the highest individual score is removed, the mean score of the remaining 7 players becomes 12 points. What was the highest score?

Answer

- A. 20
- B. 24
- C. 32
- D. 36

Question ID: 4ff597db

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard ■■■■■

Question

The mean amount of time that the 20 employees of a construction company have worked for the company is 6.7 years. After one of the employees leaves the company, the mean amount of time that the remaining employees have worked for the company is reduced to 6.25 years. How many years did the employee who left the company work for the company?

Answer

- A. 0.45
- B. 2.30
- C. 9.00
- D. 15.25

Question ID: 98958ae8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard ■■■■■

Question

Data set A consists of the heights of **75** objects and has a mean of **25** meters. Data set B consists of the heights of **50** objects and has a mean of **65** meters. Data set C consists of the heights of the **125** objects from data sets A and B. What is the mean, in meters, of data set C?

Question ID: 9d935bd8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

Percent of Residents Who Earned a Bachelor's Degree or Higher

State	Percent of residents
State A	21.9%
State B	27.9%
State C	25.9%
State D	19.5%
State E	30.1%
State F	36.4%
State G	35.5%

A survey was given to residents of all 50 states asking if they had earned a bachelor's degree or higher. The results from 7 of the states are given in the table above. The median percent of residents who earned a bachelor's degree or higher for all 50 states was 26.95%. What is the difference between the median percent of residents who earned a bachelor's degree or higher for these 7 states and the median for all 50 states?

Answer

- A. 0.05%
- B. 0.95%
- C. 1.22%
- D. 7.45%

Question ID: 54d93874

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard ■■■■■

Question

	Masses (kilograms)					
Andrew	2.4	2.5	3.6	3.1	2.5	2.7
Maria	x	3.1	2.7	2.9	3.3	2.8

Andrew and Maria each collected six rocks, and the masses of the rocks are shown in the table above. The mean of the masses of the rocks Maria collected is 0.1 kilogram greater than the mean of the masses of the rocks Andrew collected. What is the value of x ?

Question ID: 94237701

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

For a certain computer game, individuals receive an integer score that ranges from 2 through 10. The table below shows the frequency distribution of the scores of the 9 players in group A and the 11 players in group B.

Score	Score Frequencies	
	Group A	Group B
2	1	0
3	1	0
4	2	0
5	1	4
6	3	2
7	0	0
8	0	2
9	1	1
10	0	2
Total	9	11

The median of the scores for group B is how much greater than the median of the scores for group A?

Question ID: cf2be18d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

$a, 26, 29, b, 31, 47, c$

For the given data set, the data values are listed in ascending order, where a , b , and c are constants. For this data set, the mean is **36**, the median is **29**, and the range is **72**. What is the value of c ?

Answer

- A. **54**
- B. **72**
- C. **81**
- D. **98**

26. Near the end of a city-council livestream, the host invited viewers to answer a poll on the show's website: "Do you support the new parking rule discussed tonight?" The host later reported that 31% answered "Yes" and 66% answered "No." Which of the following best explains why the results are unlikely to represent the residents of the city?

- A) The percentages do not add up to 100%, so any possible conclusions from the poll are invalid.
- B) Those who responded to the poll were not a random sample of the residents of the city.
- C) There were not 50% "Yes" responses and 50% "No" responses.
- D) The show did not allow viewers enough time to respond to the poll.

27. A botanist wants to know whether a new fertilizer changes tomato yield. She has 40 plants of the same variety. She assigns 20 plants at random to the new fertilizer and 20 to the old fertilizer, grows them in the same greenhouse, and compares the mean yield. Which statement is true?

- A) This is an observational study, so a difference in mean yield cannot be used to claim the fertilizer caused the change.
- B) This is an experiment with random assignment, so a difference in mean yield can be used to claim the fertilizer caused the change.
- C) The results can be generalized to all plants of every species, because 40 is a large sample.
- D) The study is biased because the two groups do not have the same number of plants.

28. A clinic director selected 80 patients at random from **all** patients at the clinic. The table shows only the 35 patients in that sample who had at least 4 visits last year.

	4 to 6 visits	7 or more visits	Total
Improved	12	9	21
Did not improve	8	6	14
Total	20	15	35

Which of the following is the largest population to which the results of the study can be generalized?

- A) All clinic patients who improved
- B) The 80 patients in the sample
- C) The 35 patients in the sample who had at least 4 visits
- D) All patients at the clinic

29. A school counselor notices that students who already eat breakfast in the cafeteria have higher math-quiz averages than students who do not. She did not assign anyone to eat or skip breakfast.

Which statement is true?

A) This is an experiment, so she can conclude that eating breakfast caused the higher averages.

B) This is an observational study, so she cannot conclude that eating breakfast caused the higher averages.

C) The finding can be generalized to all teenagers in the country, because a school is a random sample of the country.

D) The two groups must have the same number of students, or the comparison is invalid.

30. Two random samples of votes on the same city measure were selected from the same population of voters. The margins of error were calculated the same way.

Sample	Percent in favor	Margin of error
A	54%	4.8%
B	50%	1.5%

Which of the following is the most appropriate reason that the margin of error for sample A is greater than the margin of error for sample B?

- A) Sample A had a higher percent of favorable responses.
- B) Sample A had a larger sample size.
- C) Sample A had a smaller sample size.
- D) Sample A had fewer ballots that could not be recorded.

31. A random sample of residents in a town shows that 38% support a new park, with a margin of error of 3%. Which conclusion is most appropriate?

- A) Exactly 38% of all residents in the town support the park.
- B) The true percent of all residents who support the park is likely between 35% and 41%.
- C) The park will be approved.
- D) The sample is biased because the result is not 50% to 50%.

32. Two random samples are taken from the same city. Sample P has 80 people. Sample Q has 800 people. The same method is used to compute each margin of error. Which sample is likely to have the **smaller** margin of error?

- A) Sample P
- B) Sample Q
- C) The two margins of error must be equal.
- D) It cannot be determined without the sample percents.

33. A random sample of voters shows 52% support, with a margin of error of 4%. Which conclusion is most appropriate?

- A) Exactly 52% of all voters support the candidate.
- B) The true support is likely between 48% and 56%.
- C) The candidate will win.
- D) The sample is biased because the result is not 50–50.

34. A researcher asks only students in the honors hallway whether they like a new schedule, then claims the **whole school** supports it. The main problem with this claim is:

- A) The mean is undefined.
- B) The sample is not representative of the school.
- C) The margin of error must be 0.
- D) This is a well-designed experiment.

35. A poll is 61% with margin of error 5%. Which statement is **not** justified?

- A) A reasonable interval for the true percent is 56% to 66%.
- B) More than half of the population might favor the idea.
- C) Exactly 61% of the whole population favors the idea.
- D) A larger random sample would likely give a smaller margin of error.

36. (SPR) The frequency table summarizes a data set.

Data value	Frequency
4	9
5	12
6	8
7	0
8	5
9	11

What is the maximum data value in the data set?

37. The frequency table summarizes a data set.

Data value	Frequency
10	2
20	20
30	8
40	1

What is the maximum frequency value in the data set?

- A) 20
- B) 40
- C) 400
- D) 61

38. (SPR) The frequency table summarizes a data set.

Data value	Frequency
2	4
5	15
8	7
11	2
14	6
17	0
18	3

What is the maximum data value in the data set?

39. Data set A is 8, 10, 10, 11, 15. Data set B is created by adding 7 to each value in data set A. Which statement is true?

- A) The median of B equals the median of A, and the range of B equals the range of A.
- B) The median of B equals the median of A, and the range of B is greater than the range of A.
- C) The median of B is greater than the median of A, and the range of B equals the range of A.
- D) The median of B is greater than the median of A, and the range of B is greater than the range of A.

40. (SPR) Two data sets of 25 integers each are summarized by the same frequencies.

Interval	Frequency
$0 \leq x < 10$	5
$10 \leq x < 20$	6
$20 \leq x < 30$	7
$30 \leq x < 40$	7

The first interval is the integers greater than or equal to 0 and less than 10, and so on. What is the **smallest** possible difference between the mean of data set A and the mean of data set B ?

41. (SPR) Two data sets of 23 integers each are summarized by the same frequencies.

Interval	Frequency
$20 \leq x < 30$	3
$30 \leq x < 40$	4
$40 \leq x < 50$	7
$50 \leq x < 60$	9

The first interval is the integers greater than or equal to 20 and less than 30, and so on. What is the **greatest** possible difference between the mean of data set A and the mean of data set B ?